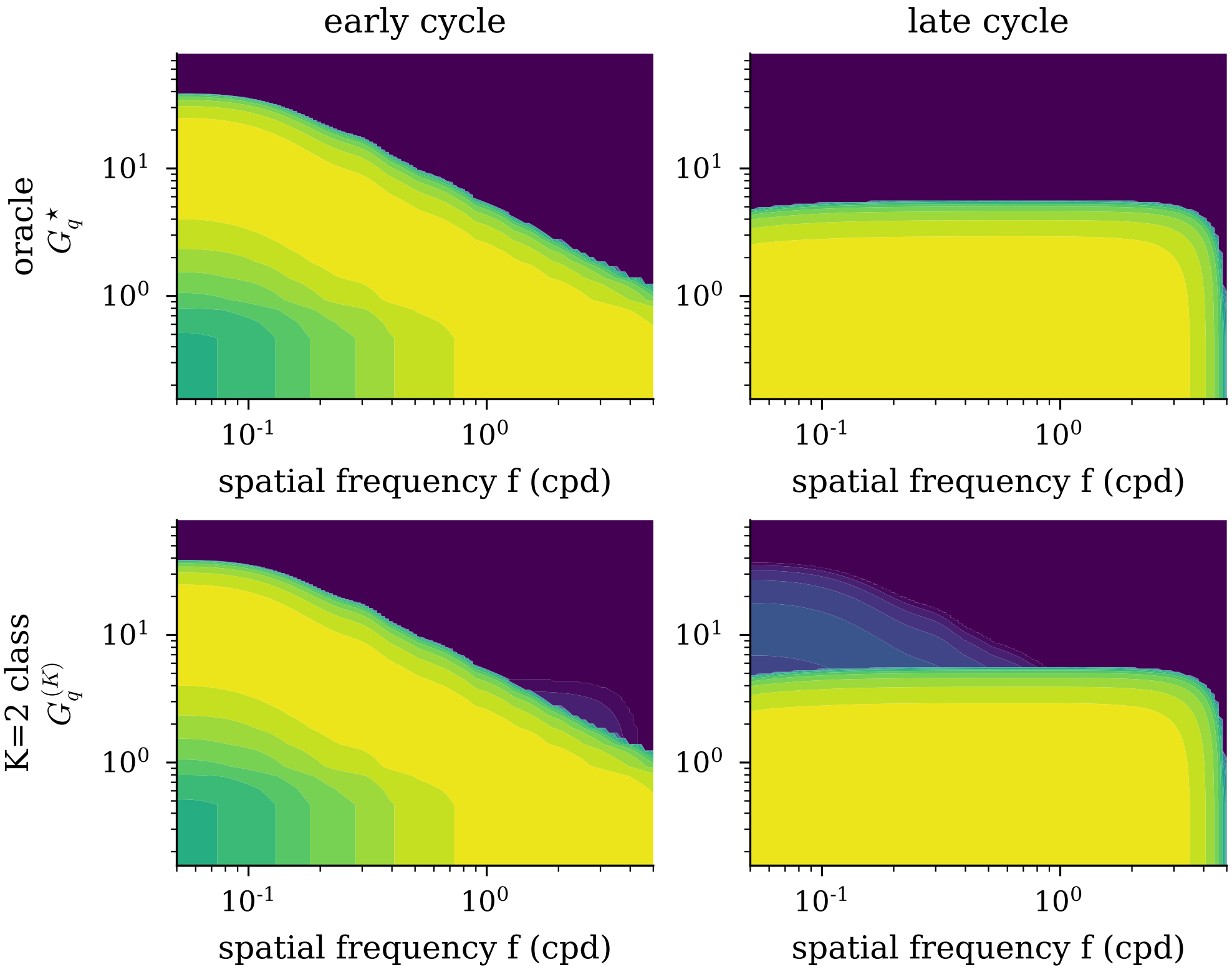Fully adaptive optima versus class-constrained filters